

# Sheraz Hassan

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## EDUCATION

### Georgia Institute of Technology

*Masters of Science in Computer Science*

Atlanta, GA

Aug 2023 – Dec 2026

**Relevant Coursework:** Graduate Algorithms, Machine Learning, Deep Learning, Ubiquitous Computing, Computer Vision, Data Visualization

### Lahore University of Management Sciences

*Bachelors of Science in Electrical Engineering*

Lahore, Pakistan

Sep 2017 – May 2021

**Relevant Coursework:** Data Structures, Artificial Intelligence, Applied Probability, Introduction to Data Science, Embedded Systems

## EXPERIENCE

### Software Engineering Intern

May 2026 – Jul 2026

*Tractian*

Atlanta, GA, USA

- Designed and shipped an internal developer platform on Kubernetes, using Terraform, ArgoCD, and Helm to expose **REST APIs** for deployment, Vault-based secret injection, and **Datadog**-monitored health checks, reducing deployment time by **over 70%**.
- Built the platform's AI layer, an autonomous provisioning agent on the **Claude Agent SDK** that drafts infrastructure code, and a production **RAG assistant** over **80+** internal PDFs and training videos with a **weekly** refresh pipeline to prevent stale answers, combining Whisper transcription, pgvector retrieval, and cross-encoder reranking, adopted by **100+** employees after **winning the company hackathon**.

### Graduate Research Assistant

Aug 2023 – May 2026

*Georgia Institute of Technology*

Atlanta, GA, USA

- Built a full-stack ML deployment platform using React and FastAPI that benchmarks **TinyML** models against **9 biomedical baselines** across **15 REST endpoints**, then serializes int8-quantized models into TensorFlow Lite Micro's FlatBuffer format and cross-compiles firmware for **3 MCU architectures**, flashing microcontrollers directly from the browser over WebSerial.
- Engineered a real-time laser speckle-imaging and vibrometry pipeline in Python on a Raspberry Pi Zero 2, extracting heart rate, respiration, and perfusion from optical signals and resolving skin-surface features from **161 to 664 micrometers**, validated against reference sensors, producing a **first-author publication** at IEEE-BSN 2025.
- Built the embedded firmware in **C** and **sensor-fusion** software for a multi-sensor vaping-detection device, integrating magnetometer, touch, and IMU inputs to estimate puff timing and intensity in real time, validated across **750+** real-world puffs and published at ACM-CHASE.

### Machine Learning Engineer

Jun 2021 – Jul 2023

*Smart Data Systems and Applications Laboratory*

Lahore, Pakistan

- Engineered and trained a CNN-LSTM demand-forecasting model in PyTorch, fusing **8 external spatiotemporal variables** including time, weather, and location to improve long-term rolling-window accuracy by **12%** over state-of-the-art baselines and shipping the model to **Careem** for production use.
- Developed and **deployed on wearable hardware** a **20K**-parameter ELM-Kalman heart-rate estimator, **92%** smaller than prior state of the art at **0.59 BPM** accuracy, then ran an adversarial ML evaluation with two novel attacks exposing model degradation.

## PROJECTS

### PodSentinel: Agentic Kubernetes Incident-Response Platform

- Built an agentic RAG system that diagnoses and remediates live Kubernetes incidents, combining a **QLoRA**-fine-tuned Qwen2.5-3B model, a custom **Go MCP** server for real-time cluster queries, and a **Go** operator that remediates only after human approval. Deployed across **5 microservices** via a self-hosted **GitHub Actions**, **Helm**, and **ArgoCD** pipeline to a live public endpoint on **AWS** and **Cloudflare**.

### EdgeGuard: On-Device Entry System

- Distilled a MobileNetV2 student model to **84% accuracy in under 200 KB** and deployed a DS-CNN wake-word listener at **95% accuracy in a 46 KB footprint** on an Arduino Nano 33 BLE.

### PlatePerfect: Computer Vision QA Platform

- Fine-tuned YOLOv8-nano on **10,000+ images** to segment **101 food classes** in real time, deploying an end-to-end visual-consistency QA system at **68% precision** for a restaurant.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Golang, JavaScript/TypeScript, MATLAB, R, SQL, Bash

**Machine Learning and AI:** PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV, Hugging Face, LangChain, LangGraph, Claude Agent SDK, Model Context Protocol (MCP), TensorFlow Lite Micro

**Web and Backend:** Next.js, React, Node.js, FastAPI, REST APIs, PostgreSQL, MySQL, Redis, Flutter

**Cloud and Infrastructure:** AWS, Oracle Cloud Infrastructure, Kubernetes, Docker, Helm, Terraform, ArgoCD, GitHub Actions, Bazel, Linux, Datadog, MLflow

## TEACHING EXPERIENCE

### Georgia Institute of Technology

Atlanta, GA

*Graduate Teaching Assistant - Artificial Intelligence*

Jan 2024 – Present

- Supported **600+ students** under Dr. Thad Starner and Dr. Thomas Ploetz, authoring graduate-level assignments and providing technical mentorship.

## PUBLICATIONS

**SkinSpex: A Portable Speckle Imaging Prototype for Multiple Skin Biomarker Detection**

IEEE-BSN, 2025

**PuffEM: An E-cigarette Sleeve for Estimating User Nicotine Intake**

ACM-CHASE, 2025

**STEF-DHNet: Spatiotemporal External Factors Based Deep Hybrid Network for Taxi Demand Prediction**

arXiv, 2023